

A040 – The Fractal Correction: $D_f = 3 - \xi$ and K_{frak}

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.1 Purpose

The geometry of A030 is described as smooth, but not intended to be smooth. FFGFT assumes that the effective structure carries a dimension deficit: it does not behave like an exactly three-dimensional object, but like one with

$$D_f = 3 - \xi.$$

This document states what follows from this, and – the actual purpose – traces the correction factor K_{frak} back to a purely internal reason.

Why both forms appear here: the closure comma gives the *internal* reading of the factor 100, the scale flow its *derivation* (below). Both describe the same number; the first shows where it sits in the geometry, the second why it is exactly 100.

.2 The Conformal Form of the Correction

[STIPULATION] The correction acts conformally on the metric:

$$g_{\mu\nu}^{(\text{frak})} = \eta_{\mu\nu}(1 + \xi f),$$

with a bounded function f on the torus.

(Note of 2 Oct. 2026 on notation: here f is a bounded function on the torus (conformal correction $g = \eta(1 + \xi f)$), not a frequency and not the base winding number $f = 1/\xi$. Cf. Doc. 190, R147.)

[B] Two properties follow immediately from this form.

First, the intervention is *metric*, not topological. A bounded, ξ -small modulation of the prefactor does not tear a manifold; it changes distances, not connections. Winding numbers and spectral structure remain untouched. This is why the fractal correction leaves everything from A030 in place.

Second, the correction is *small of order* ξ . In the limit $\xi \rightarrow 0$ it vanishes and the smooth description is exact. The idealised smooth structure is therefore not wrong, but the limiting case.

[B] Explicitly: fractal here does not mean discrete. The structure remains an uncountable continuum; it is simply not smoothly affine. One who concludes from $D_f < 3$ to a granularity in the sense of smallest building blocks has used a different concept.

.3 The Correction Factor from the Closure Comma

[B] The recursion (A050) closes in the frozen case after exactly 75 circuits, where

$$75 = \frac{1}{100\xi}, \quad \xi = \frac{4}{30000}.$$

Requiring that 75 rotations each shortened by ε close *exactly*, i.e.

$$75(1 - \varepsilon) = 74,$$

it follows that

$$\varepsilon = \frac{1}{75} = 100\xi,$$

and therefore

$$K_{\text{frak}} = 1 - 100\xi = \frac{74}{75} = 0.98\bar{6}.$$

This is the internal form: the correction factor is the ratio between the unrolled and the projected rotation. It is not a fitted value but the closure comma of a winding that falls one rotation short.

The closure comma re-expresses 100ξ ; it is the *internal* reading of the factor, not its derivation. The actual derivation of the factor 100 runs through the scale flow.

.4 Why the Factor 100

[K] The factor 100 is not stipulated but follows from the cumulative action of the sub-dimensionality $D_f = 3 - \xi$ over the entire renormalisation group run. The local deviation from dimension 3 is tiny ($\sim 0.003\%$); it becomes physically effective only *cumulatively* over the path from the Planck to the electroweak scale:

$$\ln\left(\frac{\ell_{\text{EW}}}{\ell_P}\right) \approx \ln(10^{17}) \approx 39.$$

From this run and geometric averaging over scales, an effective amplification factor of around 100 emerges: the cumulative fractal correction is not ξ , but $100\xi \approx 1.3\%$.

[K] The equivalent power form confirms this:

$$K_{\text{frak}} = \left(\frac{D_f^{\text{eff}}}{3}\right)^{D_f^{\text{eff}}/2} = 1 - 100\xi, \quad D_f^{\text{eff}} \approx 2.973.$$

The effective dimension $D_f^{\text{eff}} \approx 2.973$ is *not* the local $D_f^{\text{space}} = 2.9999$; it describes the cumulative action over the RG run. Substituting gives 0.98665 versus $1 - 100\xi = 0.98667$.

[K] And D_f^{eff} itself is uniquely determined, not free: it is the only value for which two independent derivations of the ratio m_e/m_μ – the fractal and the directly-geometric ($5\sqrt{3}/18 \cdot 10^{-2}$) – agree. From this consistency condition $D_f^{\text{eff}} \approx 2.973$ follows, and thereby $K_{\text{frak}} = 1 - 100\xi$ necessarily. The factor 100 is therefore a derived quantity.

.5 Where the Factor Appears and Where It Does Not

[B] K_{frak} concerns exclusively *absolute* values, never ratios. A ratio of two quantities at the same level is unaffected by the correction, because it cancels.

This is a sharp and testable separation, and simultaneously the most important statement of this document for everything that follows: the dimensionless ratios of the theory are independent of the fractal correction. Deviations attributable to K_{frak} can therefore only appear in the conversion to absolute values (A080), not in the geometric core.

[B] The value $74/75$ is moreover not only derived but *confirmed* in the fine-structure sector. There (A130) two closely related paths lead to the characteristic energy E_0 : the geometric mean of the lepton masses without and with the fractal correction. They differ only by K_{frak} , and the measured α fixes this factor at $0.98650 - 1.7 \cdot 10^{-4}$ from the value that follows here from the scale run. K_{frak} can therefore only have this value; this is the strongest support for $74/75$ not being freely chosen. *(Corrected on 30 Sept. 2026: previously "two separate paths ..., neither of which uses K_{frak} ; their difference is exactly $K_{\text{frak}}^{-1/2}$ " – the second path was the formula $E_0^2 = 4\sqrt{2} m_\mu / \xi^p$ with a fitted p ; it has been removed from A130. K_{frak} is fixed via the measured α .)*

.6 The Limit of Resolution

[B] The dimension deficit is a deficit, not a volume-preserving ripple. The difference is computationally accessible: for a volume-preserving modulation, the mean of f over the torus vanishes; for a deficit, it does not.

This decides with what power of ξ the correction enters first-order perturbation calculations – with nonvanishing mean as ξ^1 , otherwise only as ξ^2 . The verification script computes both cases and shows the exponents numerically.

.7 Verification Scripts

- Closure comma: $75(1 - \varepsilon) = 74 \Rightarrow \varepsilon = 1/75 = 100\xi$, with explicit countercheck that the computation assumes rather than produces the 100: `python/A_Serie_Skripte/a040_schliessungskomma.py`
- Deficit versus volume-preserving ripple: numerical determination of the ξ -power in both cases: `python/A_Serie_Skripte/a040_dimensionsdefizit.py`

.8 Open Questions

First, the closure comma shows consistency, not necessity. The cumulative RG run derives $K_{\text{frak}} = 1 - 100\xi$ from the accumulated effect of $D_f = 3 - \xi$ over 17 orders of magnitude; the factor 100 emerges as a logarithmic scale ratio. The closure comma ($1 - 100\xi$ hits the value 0.98650 determined from α to within $1.7 \cdot 10^{-4}$) *(Corrected on 1 Oct. 2026: previously "hits exactly the empirical value" – $\xi m_e m_\mu / \alpha = 0.986501$ versus $74/75 = 0.986667$, a gap of $1.66 \cdot 10^{-4}$, as in the section on the confirmation in the fine-structure sector.)* confirms the construction but does not prove it: a geometric derivation showing that exactly 100 follows and no other factor still stands. This remainder is explicitly smaller than the point as previously formulated.

Second, f is not determined. The form of the correction is stipulated, the function itself is not. Everything depending on details of f – notably the phase of its harmonics – is thereby open.

Third, K_{frak} is a single, generation-independent factor. There are indications that the correction should act generation-dependently on the mass ladder; a constant factor does not close these deviations. The point belongs substantively in A100 and is treated there.

.9 Supersedes

This document subsumes: Doc. 132 (fractal duality), Doc. 133 (fractal correction derivation), Doc. 300 (two roles of K_{frak}), Doc. 210 (winding-number corrections, insofar as they concern the factor). Incorporated: K1–K6 from the correction register, insofar as they concern the correction form.

.10 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.